SYSTEM STUDY

1. Introduction

A System Study is the process of analysing the existing system, identifying its limitations, and proposing an improved solution to address those limitations. In the domain of cybersecurity, most existing systems focus on detecting cyber threats only after they occur. However, modern cyberattacks increasingly exploit human behaviour through phishing, weak passwords, malicious URLs, and social engineering. Therefore, there is a growing need for a proactive system capable of predicting risky user behaviour before it leads to security incidents.

Cyber Sense AI is proposed as an AI-powered, website-based cybersecurity platform that predicts human cyber behaviour risks using Machine Learning and Explainable Artificial Intelligence (XAI). The system provides early warnings, explains AI predictions, and offers personalized cybersecurity coaching to improve users' cyber hygiene.

2. Existing System Study

Current cybersecurity solutions are designed to solve specific security problems individually. Examples include spam filters, antivirus software, URL reputation systems, password strength checkers, and intrusion detection systems. Although these tools are effective in their respective areas, they generally operate independently and provide only reactive protection.

Most existing systems:

- Detect phishing emails or SMS individually.
- Scan URLs for malicious content.
- Evaluate password strength.
- Monitor network traffic or malware.
- Provide generic cybersecurity awareness materials.

However, these systems do not provide a comprehensive understanding of a user's overall cyber behaviour or future cyber risks.

Limitations of Existing System

- Reactive rather than proactive.
- Focus on a single cybersecurity threat.
- Limited behavioural analysis.
- No unified cyber risk assessment.
- Lack of Explainable AI to justify predictions.
- Generic awareness programs without personalization.
- Inability to predict future cyber behaviour.

3. Proposed System Study

Cyber Sense AI introduces a proactive approach to cybersecurity by analysing multiple aspects of human cyber behaviour and predicting cyber risks before attacks occur.

The proposed system integrates multiple AI-powered modules, including:

- SMS Phishing Detection
- Email Phishing Detection
- Malicious URL Detection
- Authentication Behaviour Analysis
- Behavioural Drift Detection

Each module independently analyses its respective cyber risk using machine learning algorithms. The prediction results are then combined through a Risk Fusion Engine to calculate an Overall Cyber Risk Score representing the user's cybersecurity posture.

To improve transparency, the system incorporates Explainable AI techniques such as SHAP or LIME, allowing users to understand the reasons behind every prediction. Based on the identified risks, the platform also provides personalized cybersecurity recommendations and coaching to help users improve their online security behaviour.

4. System Objectives

The proposed system aims to:

- Predict human cyber behaviour risks before cyber incidents occur.
- Detect phishing SMS, phishing emails, and malicious URLs.
- Assess password security and authentication behaviour.
- Identify behavioural changes over time through behavioural drift detection.
- Generate an overall cyber risk score.
- Explain AI predictions using Explainable AI.
- Provide personalized cybersecurity coaching.
- Improve users' cybersecurity awareness and cyber hygiene.

5. Functional Requirements

The system should provide the following functionalities:

- User Registration and Login
- SMS Phishing Analysis
- Email Phishing Analysis
- URL Risk Analysis
- Authentication Behaviour Monitoring
- Behavioural Drift Detection
- Overall Cyber Risk Dashboard
- Explainable AI Visualization
- Personalized Recommendations
- Risk History and Reports

6. Non-Functional Requirements

Performance

- Fast AI prediction with minimal response time.

- Efficient handling of multiple users simultaneously.

Security

- Secure user authentication.
- Encrypted password storage.
- Protected user data.
- Secure communication between client and server.

Reliability

- Accurate machine learning predictions.
- Consistent system availability.
- Reliable data storage.

Scalability

- Easy addition of new cybersecurity risk modules.
- Support for larger datasets and increasing numbers of users.

Usability

- User-friendly website interface.
- Easy navigation.
- Simple risk visualization.
- Responsive design for desktop and mobile devices.

7. System Modules

The proposed system consists of the following modules:

Module 1: User Management

Handles user registration, login, profile management, and authentication.

Module 2: SMS Phishing Detection

Analyses SMS messages and predicts whether they are safe or phishing.

Module 3: Email Phishing Detection

Identifies phishing emails using AI-based text analysis.

Module 4: Malicious URL Detection

Evaluates URLs to determine whether they are legitimate or malicious.

Module 5: Authentication Behaviour Analysis

Monitors login patterns and detects unusual authentication behaviour.

Module 6: Behavioural Drift Detection

Identifies significant changes in user behaviour over time using sequential machine learning models.

Module 7: Explainable AI

Provides transparent explanations for every AI prediction using SHAP or LIME.

Module 8: Cyber Risk Dashboard

Displays individual module results, overall cyber risk score, historical trends, and security metrics.

Module 9: Personalized Cybersecurity Coaching

Generates customized recommendations and awareness activities based on the user's cyber behaviour.

8. Feasibility Study

Technical Feasibility

The project is technically feasible because it utilizes established technologies such as Python, Flask, MySQL, Scikit-learn, TensorFlow, and publicly available Kaggle datasets. Explainable AI libraries like SHAP and LIME can also be integrated without additional licensing costs.

Economic Feasibility

The proposed system is economically feasible since it relies primarily on open-source software, free development tools, and publicly available datasets, minimizing development costs.

Behavioural Feasibility

The website is designed to be simple and user-friendly, enabling students, employees, and general users to understand their cyber risks without requiring advanced cybersecurity knowledge.

9. Advantages of the Proposed System

- Proactive prediction of cyber risks before incidents occur.
- Comprehensive analysis of multiple cybersecurity risk categories.
- Integration of Explainable AI for transparent decision-making.
- Personalized cybersecurity coaching based on user behavior.
- Unified Cyber Risk Score representing overall cybersecurity posture.
- Modular architecture supporting future enhancements.
- Improved cybersecurity awareness and cyber hygiene.

10. Conclusion

The system study concludes that existing cybersecurity solutions primarily address individual threats and rely on reactive detection methods. Cyber Sense AI addresses these limitations by introducing a proactive, AI-driven platform that analyses multiple dimensions of human cyber behaviour, predicts potential cyber risks, explains AI decisions, and provides personalized guidance to improve cybersecurity practices. Its modular architecture, explainable predictions, and user-centric approach make it a scalable solution for both academic research and practical cybersecurity applications.